

Question 1 (16 Marks)**Marks**

- (a) Convert $1.\dot{6}\dot{0}$ to a fraction in its simplest terms. 2
- (b) Simplify $\frac{2^{-m} \times 3^{m-2}}{4^{m-1} \times 6^{2-m}}$ 3
- (c) Factorise fully
- (i) $8x^3 - 125$ 1
- (ii) $x^2y - 4x^2 - 9y + 36$ 2
- (d) Simplify $\frac{1}{x^2 - 1} - \frac{2}{x^2 - x}$ 2
- (e) Show that $\frac{2}{3 - \sqrt{3}} + \frac{11}{6 + \sqrt{3}}$ can be expressed as a rational number. 3
- (f) Given $6 + \sqrt{x - y} = x + y + 3\sqrt{2}$ find the values for x and y . 3

Question 2 START A NEW PAGE (14 Marks)

- (a) Solve
- (i) $\frac{x+3}{2} + \frac{x-1}{3} = \frac{x+2}{6}$. 3
- (ii) $|x+2| = 2x-5$. 3
- (iii) $2x^2 - 4x - 7 = 0$, giving your answer in the simplest exact form. 2
- (b) Solve the following inequalities
- (i) $x^2 - x - 2 > 0$. 1
- (ii) $\frac{3}{x-2} \geq 1$. 3
- (iii) $|2x-3| \geq 7$ 2

Question 3 START A NEW PAGE (10 Marks)**Marks**

- (a) Solve $x + y = 3$ and $x^2 + y^2 = 29$ simultaneously. **3**
- (b) If $f(x) = x^2 + 3x + 5$, evaluate and simplify $\frac{f(x) - f(2)}{x - 2}$, $x \neq 2$. **3**
- (c) By completing the square, find the centre and radius of the circle with the equation $x^2 - 10x + y^2 + 8y + 32 = 0$. **3**
- (d) State the natural domain of the following functions
- (i) $f(x) = \frac{1}{2x - 7}$. **1**
- (ii) $f(x) = \frac{1}{\sqrt{x + 5}}$ **1**

Question 4 START A NEW PAGE (10 Marks)

- (a) Determine whether the function $f(x) = \frac{7x}{7 + x^2}$ is odd even or neither. **2**
- (b) Sketch the following on separate number plane graphs, showing any asymptotes, vertices, intercepts or essential features.
- (i) $y = \sqrt{x + 3}$. **2**
- (ii) $y = 4 - (x + 1)^2$. **2**
- (iii) $y = |x(x - 2)(x + 2)|$. **2**
- (iv) $y = \frac{x - 2}{x - 5}$. **2**

End of Paper.

Question 1

a) Let $x = 1.\dot{6}\dot{0}$

$$x = 1.606060\dots \text{--- ①}$$

$$100x = 160.606060\dots \text{--- ②}$$

$$\text{②} - \text{①} \quad 99x = 159$$

$$x = \frac{159}{99}$$

$$= 1\frac{20}{33}$$

$$\therefore 1.\dot{6}\dot{0} = 1\frac{20}{33}$$

$$\begin{aligned}
 \text{b) } \frac{2^{-m} \times 3^{m-2}}{4^{m-1} \times 6^{2-m}} &= \frac{2^{-m} \times 3^{m-2}}{2^{2m-2} \times 2^{2-m} \times 3^{2-m}} \\
 &= \frac{2^{-m} \times 3^{m-2}}{2^m \times 3^{2-m}} \\
 &= 2^{-2m} \times 3^{2m-4}
 \end{aligned}$$

OR

$$\frac{3^{2m-4}}{2^{2m}}$$

$$\begin{aligned}
 \text{c) (i) } 8x^3 - 125 &= (2x)^3 - 5^3 \\
 &= (2x-5)(4x^2+10x+25)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } x^2y - 4x^2 - 9y + 36 &= x^2(y-4) - 9(y-4) \\
 &= (x^2-9)(y-4) \\
 &= (x-3)(x+3)(y-4)
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \frac{1}{x^2-1} - \frac{2}{x^2-x} &= \frac{1}{(x-1)(x+1)} - \frac{2}{x(x-1)} \\
 &= \frac{x-2(x+1)}{x(x-1)(x+1)} \\
 &= \frac{-x-2}{x(x-1)(x+1)} \\
 &= \frac{-(x+2)}{x(x+1)(x-1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{e) } \frac{2}{3-\sqrt{3}} &= \frac{2}{3-\sqrt{3}} \times \frac{3+\sqrt{3}}{3+\sqrt{3}} \\
 &= \frac{2(3+\sqrt{3})}{9-3} \\
 &= \frac{2(3+\sqrt{3})}{6} \\
 &= \frac{3+\sqrt{3}}{3}
 \end{aligned}$$

$$\begin{aligned}
 \frac{11}{6+\sqrt{3}} &= \frac{11}{6+\sqrt{3}} \times \frac{6-\sqrt{3}}{6-\sqrt{3}} \\
 &= \frac{11(6-\sqrt{3})}{36-3} \\
 &= \frac{11(6-\sqrt{3})}{33} \\
 &= \frac{6-\sqrt{3}}{3}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{2}{3-\sqrt{3}} + \frac{11}{6+\sqrt{3}} &= \frac{3+\sqrt{3}+6-\sqrt{3}}{3} \\
 &= \frac{9}{3}
 \end{aligned}$$

$$= 3$$

which is rational.

$$\begin{aligned}
 \text{f) } 6 + \sqrt{x-y} &= x+y + 3\sqrt{2} \\
 &= x+y + \sqrt{18}
 \end{aligned}$$

$$\therefore x+y = 6 \text{ --- ①}$$

$$x-y = 18 \text{ --- ②}$$

$$\begin{aligned}
 \text{①} + \text{②} \quad 2x &= 24 \\
 x &= 12
 \end{aligned}$$

$$\begin{aligned}
 \text{Sub ①} \quad x+y &= 6 \\
 12+y &= 6 \\
 y &= -6
 \end{aligned}$$

$$\therefore x = 12, y = -6$$

Question 2

$$a) i) \frac{x+3}{2} + \frac{x-1}{3} = \frac{x+2}{6}$$

$$3(x+3) + 2(x-1) = x+2$$

$$3x+9 + 2x-2 = x+2$$

$$5x+7 = x+2$$

$$4x = -5$$

$$x = -\frac{5}{4}$$

$$ii) |x+2| = 2x-5$$

$$x+2 = 2x-5$$

$$-x = -7$$

$$x = 7$$

$$x+2 = -(2x-5)$$

$$= -2x+5$$

$$x = 1$$

CHECK

$$LHS = |x+2|$$

$$= |7+2|$$

$$= 9$$

$$RHS = 2x-5$$

$$= 2(7)-5$$

$$= 9$$

$$= LHS$$

$$LHS = |x+2|$$

$$= |1+2|$$

$$= 3$$

$$RHS = 2x-5$$

$$= 2(1)-5$$

$$= -3$$

$$LHS \neq RHS$$

$\therefore x=7$ is a

solution

$\therefore x=1$ is

not a solution

$\therefore x=7$ is the only solution

$$iii) 2x^2 - 4x - 7 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{(-4)^2 - 4 \times 2 \times -7}}{2(2)}$$

$$= \frac{4 \pm \sqrt{12}}{4}$$

$$= \frac{4 \pm 6\sqrt{2}}{4}$$

$$= \frac{2(2 \pm 3\sqrt{2})}{4}$$

$$\therefore x = \frac{2 \pm 3\sqrt{2}}{2}$$

$$b) i) x^2 - x - 2 > 0$$

$$(x-2)(x+1) > 0$$

$$x < -1 \quad x > 2$$



$$ii) \frac{3}{x-2} \geq 1, \quad x \neq 2$$

$$(x-2)^2 \times \frac{3}{(x-2)} \geq 1 \quad (x-2)^2$$

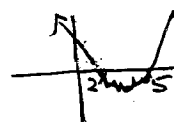
$$3(x-2) \geq (x-2)^2$$

$$(x-2)^2 - 3(x-2) \leq 0$$

$$(x-2)[(x-2)-3] \leq 0$$

$$(x-2)(x-5) \leq 0$$

$$2 < x \leq 5$$



$$iii) |2x-3| \geq 7$$

$$2x-3 \leq -7$$

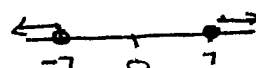
$$2x \leq -4$$

$$x \leq -2$$

$$2x-3 \geq 7$$

$$2x \geq 10$$

$$x \geq 5$$



Question 3

$$a) \quad x+y = 3 \quad \text{--- ①}$$

$$x^2 + y^2 = 29 \quad \text{--- ②}$$

rearrange ①

$$y = 3-x \quad \text{--- ③}$$

sub ③ in ②

$$x^2 + y^2 = 29$$

$$x^2 + (3-x)^2 = 29$$

$$x^2 + 9 - 6x + x^2 = 29$$

$$2x^2 - 6x - 20 = 0$$

$$x^2 - 3x - 10 = 0$$

$$(x-5)(x+2) = 0$$

$$x = 5, -2$$

$$\text{when } x=5$$

$$\text{when } x=-2$$

$$\text{sub ① } x+y=3$$

$$\text{sub ② } x+y=3$$

$$5+y=3$$

$$-2+y=3$$

$$y=-2$$

$$y=5$$

$$\therefore \left. \begin{matrix} x=5 \\ y=-2 \end{matrix} \right\} \quad \left. \begin{matrix} x=-2 \\ y=5 \end{matrix} \right\}$$

$$b) f(x) = x^2 + 3x + 5$$

$$f(2) = 2^2 + 3(2) + 5$$

$$= 15$$

$$\therefore \frac{f(x) - f(2)}{x-2} = \frac{x^2 + 3x + 5 - 15}{x-2}$$

$$= \frac{x^2 + 3x - 10}{x-2}$$

$$= \frac{(x+5)(x-2)}{(x-2)}$$

$$\therefore \frac{f(x) - f(2)}{x-2} = x+5$$

$$c) x^2 - 10x + y^2 + 8y + 32 = 0$$

$$x^2 - 10x + \left(\frac{-10}{2}\right)^2 + y^2 + 8y + \left(\frac{8}{2}\right)^2 = -32 + 25 + 16$$

$$(x-5)^2 + (y+4)^2 = 9$$

$$(x-5)^2 + (y+4)^2 = 3^2$$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$\therefore \text{centre is } (5, -4)$$

$$\text{radius is } 3$$

$$d) \text{ i) } f(x) = \frac{1}{2x-7}$$

$$2x-7 \neq 0$$

$$2x \neq 7$$

$$x \neq 3\frac{1}{2}$$

$$\therefore \text{domain } x \neq 3\frac{1}{2}$$

$$\text{ii) } f(x) = \frac{1}{\sqrt{x+5}}$$

$$x+5 > 0$$

$$x > -5$$

$$\therefore \text{domain } x > -5$$

Question 4

$$a) f(x) = \frac{7x}{7+x^2}$$

$$f(-x) = \frac{7(-x)}{7+(-x)^2}$$

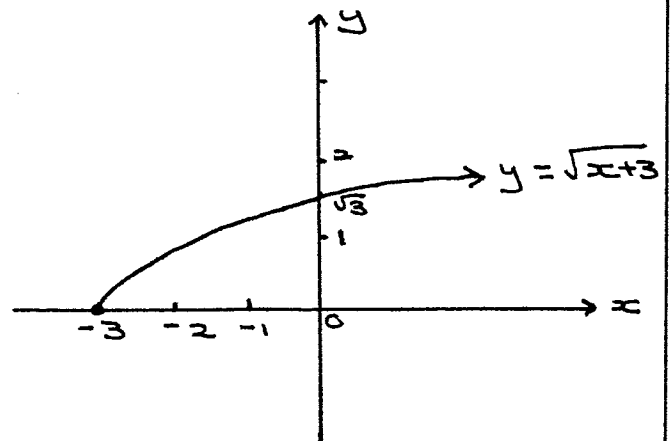
$$= \frac{-7x}{7+x^2}$$

$$-f(x) = -\left(\frac{7x}{7+x^2}\right)$$

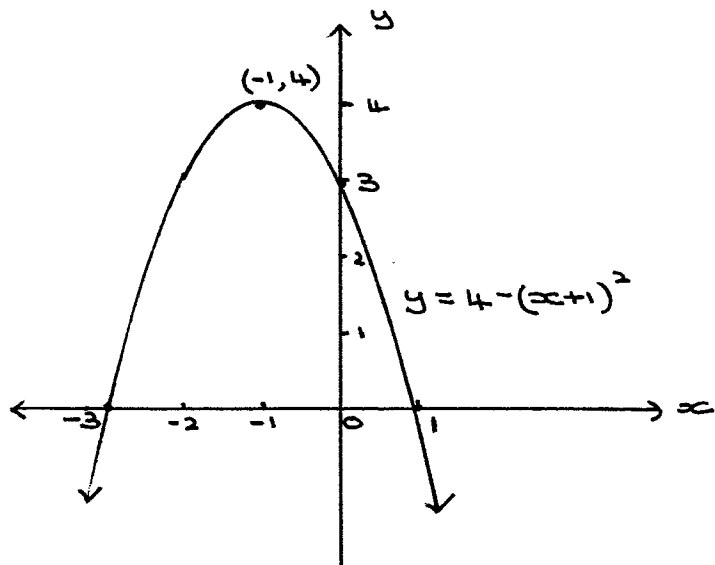
$$= \frac{-7x}{7+x^2}$$

As $f(-x) = -f(x)$ the function is odd.

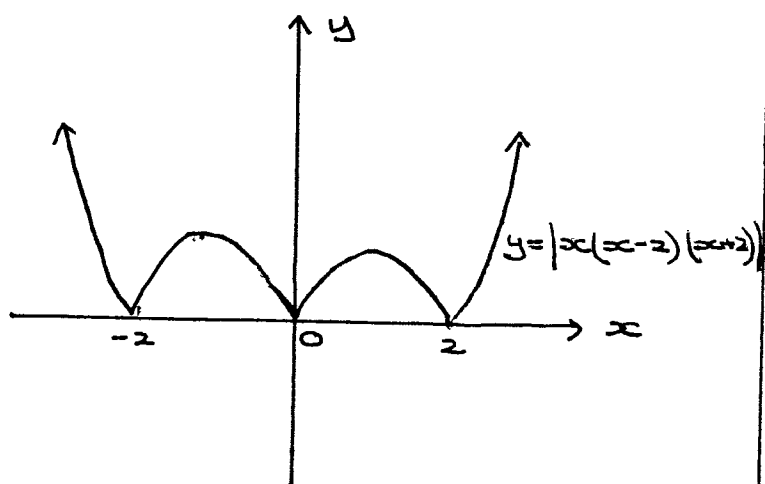
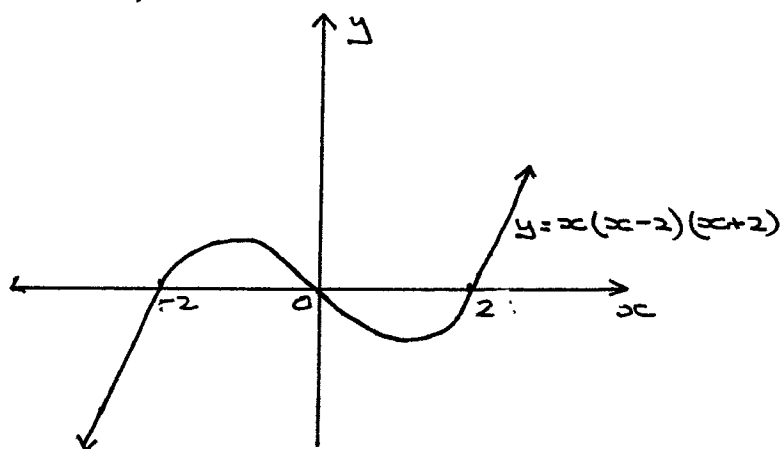
$$b) \text{ i) } y = \sqrt{x+3}$$



ii) $y = 4 - (x+1)^2$



iii) $y = |x(x-2)(x+2)|$



(iv) $y = \frac{x-2}{x-5}$

$$= \frac{x-5}{x-5} + \frac{3}{x-5}$$

$$= 1 + \frac{3}{x-5}$$

when $x=0$

$$y = \frac{0-2}{0-5}$$

$$= \frac{2}{5}$$

$\therefore$ y intercept $(0, \frac{2}{5})$

when $y=0$

$$y = \frac{x-2}{x-5}$$

$$0 = \frac{x-2}{x-5}$$

$$x-2=0$$

$$x=2$$

$\therefore$ x intercept $(2, 0)$

